Vinayak Kamat

AI/ML Engineer | LLM Systems and Backend

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vinayakkamatcodes

Summary

AI/ML Engineer with 1 year 8 months at KAN Innovations building production ML and LLM systems in Python. Hands-on with FastAPI services, RAG pipelines, ONNX inference optimization, and Docker-based AWS deployments (Lambda, ECR, S3, DynamoDB). Built Aethelgard, a real-time LLM hallucination detector with Datadog monitoring, and nanoMind, an offline on-device LLM app. MSc candidate in Financial Engineering.

Skills

Languages: Python, JavaScript, C++, Dart

Backend and Cloud: FastAPI, Flask, REST APIs, Docker, AWS (Lambda, ECR, S3, DynamoDB)

LLM and RAG: RAG pipelines, LangChain, LlamaIndex, Chroma, prompt engineering, hallucination detection, llama.cpp (GGUF)

ML and Inference: PyTorch, ONNX, Transformers, spaCy, OpenCV, MLflow

Observability: Datadog (metrics streaming), experiment tracking, model versioning

Experience

KAN Innovations, Junior AI/ML Engineer — Mumbai, Maharashtra
May 2025 – present
1 year 6 months

- Built and evaluated end-to-end RAG pipelines covering retrieval, prompting, and generation
- Deployed ML inference services on AWS Lambda using Docker images on ECR, with S3 and DynamoDB for storage and state
- Converted PyTorch models to ONNX to speed up inference in production
- Improved model calibration so prediction confidence better reflects real accuracy, making outputs more reliable in production
- Tracked experiments and versioned models with MLflow for reproducible workflows

KAN Innovations, ML Engineer Intern — Mumbai, Maharashtra
Feb 2025 – May 2025
3 months

- Built a PyTorch ulcer detection model reaching about 85% accuracy
- Improved the performance of an eye posture detection model

Projects

Aethelgard

Real-time AI governance gateway that analyzes LLM outputs for hallucinations and risk.

- Built a FastAPI gateway that analyzes LLM responses in real time, using a secondary "surgeon" model for semantic risk scoring
- Streamed risk metrics to Datadog for monitoring of LLM output quality
- Designed automated hallucination detection logic to support safer LLM deployments

nanoMind

Android chat app that runs LLMs fully offline with llama.cpp and Kotlin.

- Enabled on-device GGUF model inference with real-time token streaming using Kotlin Coroutines and Jetpack Compose
- Optimized model loading with ARM-specific performance improvements

Financial AI Agent (Demo)
Prototype agent for financial data retrieval and reasoning in Python.

- Built modular retrieval and reasoning pipelines with query routing and response refinement

Education

MSc **WorldQuant University**, Financial Engineering — Remote, Washington DC
Sept 2025 – May 2027

B.Tech **Punjab Technical University**, Computer Science — Punjab, India
Sept 2021 – May 2025

Achievements

Qualified the Graduate Aptitude Test in Engineering (GATE)